



MTM4681 - Mobile Programming

Fall 2026

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Engineering

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Key Information

- Thursday, 9:00 am - 11:50 am @ KBM - 211
- Bologna : MTM4681 Mobile Programming - YTU Bologna
- Fall 2026 Form: Weeks 1-7 + 9-13 teaching, **Week 8 Midterm**
- Attendance: Required by **REGULATIONS!!!**



Grading Breakdown (100%)

Midterm 30% + Final 40% + Project 30% = 100%

- Mid-Term (30%) and Final Exam (40%) — Face to Face, room TBA
- **Project 30% = Milestone Report 10% + Final Demo + Code 20%**
- **Recap Bonus: up to +10% *on the Project grade*** (individual, capped at 100)

Example: Project 80/100 + 8 bonus = 88/100 → 26.4/30 to final grade.



Semester Project — Real-World Problem with Flutter

Goal: Not a toy calculator. Identify a real user, real pain, and ship a working Flutter app with **local + cloud data, navigation, and clean MVVM**.

- **Team size: 3 to 5 members**

Math-engineering suitable examples:

1. Campus shuttle / classroom finder with GPS + timetable
2. Expense-split + budget forecast with charts
3. Patient medication tracker with reminders + Firestore sync
4. Earthquake safe-zone + emergency kit app for Istanbul
5. Study planner with streaks, local notifications, real-time leaderboard
6. Numerical solver mobile: plot, root-find, linear systems with history sync

Why is the number of a team 3 to 5?

1. **Roles map to Flutter reality:** UI/widgets, navigation/state, network/database, sensors/testing, docs. <3 can't cover this in 12 weeks.
2. **No free-riding:** >5 = coordination cost $>$ coding benefit, unfair peer review. 3–5 keeps everyone code-visible on GitHub.
3. **Fair workload:** ~ 20 h project / person. 3-person = MVP scope, 5-person = must add real-time + sensor + multilingual to get full points.

Team list due Week 3.

Milestone 10%

Submit PDF + 1 Figma/paper mockup:

1. Problem + user: who suffers, how often, evidence (1 stat/interview)
2. Why important: why existing apps fail
3. Features: Must-have 5–6 mapped to course topics (navigation, API/JSON, local DB, Firebase, animation, MVVM)
4. Architecture: screens map, data model, MVVM diagram, Provider/Riverpod choice
5. Plan: weekly tasks per member, risks

Submit: Private GitHub repo — share link + screenshot (SS) with professor at MS1 submission.

Project Final 20%

1. 10-min demo: real phone, offline/online, no slides-only
2. Private GitHub repo: clean commits per member, README with setup, pubspec, screenshots
3. APK / build + 2-min video backup
4. Peer contribution form

Submit: Private GitHub repo — share link + screenshot (SS) with professor at Final submission.

Rubric 20pts: Working app 8, Flutter quality (navigation, state, async, error handling) 6, Cloud/local data 3, Presentation + Q&A 3.

Recap Quizzes — Extra up to 10%

- 5-min individual question at start of class from last week — e.g. `Future` vs `Stream`, `Navigator.push`, Firestore rules
- No make-up. Best scores added as bonus to *your* project portion only, not to midterm/final
- Rewards attendance without changing official 30/40/30 formula

MTM4681 - Readiness Survey

https://docs.google.com/forms/d/e/1FAIpQLSeZktvCwdnPSH7dtbq_MKarB2qLdd7uT1FUzBIU4d-jBZDL-A/viewform?usp=publish-editor